

COSMIC Full Function Points (COSMIC FFP) in Software Project Management

1. What is COSMIC?

COSMIC stands for **Common Software Measurement International Consortium**.

COSMIC Full Function Points (COSMIC FFP) is a **software sizing method** used to measure the **functional size of software**.

It is especially useful for estimating:

- Software effort
- Development cost
- Development time
- Productivity
- Project size

Simple Definition

COSMIC Function Points is a standardized method for measuring the functional size of software by counting the data movements that occur across the software's functional processes.

The important idea is:

COSMIC measures what the software does in terms of data movement, rather than counting lines of code.

2. Why Do We Need COSMIC?

Suppose a client says:

"Build an online banking application."

Before development, the project manager needs to estimate:

- How large is the software?
- How much effort will be required?
- How many developers are needed?
- How much will development cost?
- How long will development take?

Counting **Lines of Code (LOC)** is difficult before coding starts.

For example, we don't know whether the application will require:

20,000 or 50,000 lines of Java/Python code.

COSMIC provides another approach.

Instead of asking:

"How many lines of code will we write?"

we ask:

"What data does the software receive, produce, read, and write?"

3. COSMIC Measures Functional Size

COSMIC measures **functional size**.

Functional size means:

The amount of functionality that software provides to its users.

For example, an ATM provides functions such as:

- Enter PIN
- Check balance
- Withdraw money
- Deposit money
- Print receipt

COSMIC looks at the **data movements involved in these functions**.

4. Basic COSMIC Concept

COSMIC divides software functionality into:

Functional Processes

A **functional process** is a set of activities that together achieve a meaningful task for a user.

Examples:

- Login
- Withdraw cash
- Create account
- Place order
- Generate invoice

Within these processes, data moves between:

- User and software
- Software and persistent storage
- Software and external systems

COSMIC measures these **data movements**.

5. Four Types of COSMIC Data Movements

This is the **most important concept** for exams.

COSMIC identifies **four types of data movements**:

1. Entry (E)

2. Exit (X)

3. Read (R)

4. Write (W)

Remember:

E → X → R → W

Let's understand each.

6. Entry (E)

An **Entry** occurs when data moves **from a user or external system into the software**.

Example:

A user enters:

Username
Password

into a login application.

The movement:

User → Software

is an **Entry**.

Examples:

- Login details
- Customer registration data
- Payment request
- Search query
- ATM PIN

7. Exit (X)

An **Exit** occurs when data moves **from the software to a user or external system**.

Example:

The software sends:

Login Successful

to the user.

Software → User

This is an **Exit**.

Examples:

- Bank statement
- Search results
- Invoice
- Error message
- Payment confirmation

8. Read (R)

A **Read** occurs when the software retrieves data from **persistent storage**.

Database → Software

Example:

During login:

Software → Database

asks for the stored password information.

Database → Software

returns the user record.

The second movement is a **Read**.

Examples:

- Reading customer details.
 - Reading product information.
 - Reading account balance.
 - Reading employee records.
-

9. Write (W)

A **Write** occurs when software stores or updates data in persistent storage.

Software → Database

Example:

When a customer creates an account:

Software → Customer Database

The customer information is stored.

This is a **Write**.

Examples:

- Creating a new customer record.
 - Updating account balance.
 - Saving an order.
 - Storing payment information.
-

10. The COSMIC Measurement Unit

Each data movement is counted as:

1 COSMIC Function Point (CFP)

Therefore:

Data Movement COSMIC CFP

Entry (E)	1 CFP
Exit (X)	1 CFP
Read (R)	1 CFP
Write (W)	1 CFP

Formula

COSMIC Functional Size = $E + X + R + W$

This is the most important formula for basic COSMIC questions.

11. Simple Example: Login System

Suppose a login process works like this:

```
User
↓
Username + Password
↓
Software
↓
Read User Database
↓
Verify Credentials
↓
Software
↓
Login Result
↓
User
```

Now identify the movements.

Step 1 — Entry

Username and password:

User → Software

= **1 Entry**

Step 2 — Read

Software retrieves user information:

Database → Software

= **1 Read**

Step 3 — Exit

Login result:

Software → User

= **1 Exit**

Therefore:

$CFP = E + R + X$ $CFP = 1 + 1 + 1$ **3 CFP**

So this functional process has a size of **3 COSMIC Function Points**.

12. Example: Online Shopping — Place Order

Suppose a customer places an order.

Data movements:

1. Customer sends order information → **Entry**
2. System reads product information → **Read**
3. System reads customer information → **Read**
4. System writes the new order → **Write**
5. System sends order confirmation → **Exit**

Therefore:

Movement Count

Entry	1
Read	2
Write	1
Exit	1

Total:

$$1+2+1+1=5$$

Functional Size:

5 CFP

13. COSMIC Software Model

COSMIC considers three major aspects:

1. Functional Users

Entities interacting with the software.

Examples:

- Customer
- Administrator
- ATM
- External payment system

2. Functional Processes

Tasks performed by the software.

Examples:

- Login
- Withdraw money
- Place order
- Generate report

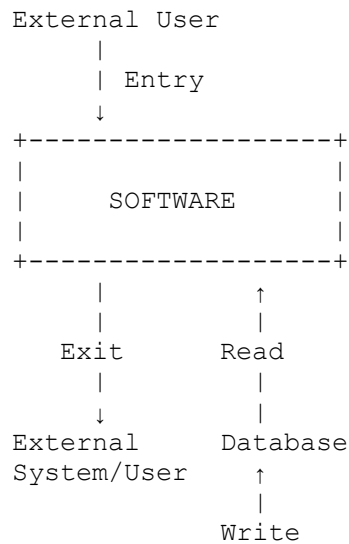
3. Data Movements

Movement of data:

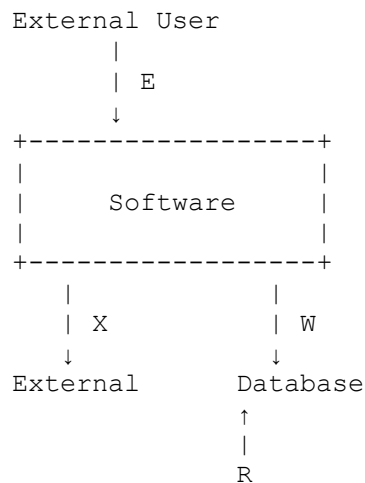
Entry → Exit → Read → Write

14. COSMIC Generic Software Model

A simple representation is:



More accurately:



15. COSMIC Measurement Process

COSMIC measurement generally involves four major activities.

Step 1: Define Measurement Purpose

Determine **why** you are measuring the software.

For example:

We want to estimate development effort and cost.

Step 2: Identify the Software Boundary

Determine what is inside and outside the software being measured.

Example:

Customer → [Online Shopping System] → Payment Gateway

The shopping system is the software being measured.

Step 3: Identify Functional Processes

Identify the major functional processes.

For an online shopping system:

- Login
 - Search Product
 - Add to Cart
 - Place Order
 - Make Payment
 - Track Order
-

Step 4: Identify Data Movements

For each functional process, identify:

- Entry
- Exit
- Read
- Write

Then count them.

16. Detailed Example: ATM Withdrawal

Consider an ATM withdrawal process.

Step 1

Customer enters card/PIN/withdrawal request.

Customer → ATM Software

Entry = 1

Step 2

ATM software reads customer/account information.

Database → ATM Software

Read = 1

Step 3

ATM software updates account balance.

ATM Software → Database

Write = 1

Step 4

ATM provides cash/transaction information to customer.

ATM Software → Customer

Exit = 1

Total

CFP=1E+1R+1W+1X 4 CFP

17. COSMIC vs Traditional Function Points

Students often confuse **COSMIC Function Points** with traditional **Function Point Analysis (FPA)**.

Traditional Function Points	COSMIC Function Points
Uses 5 function types	Uses 4 data movement types
EI, EO, EQ, ILF, EIF	Entry, Exit, Read, Write
Uses complexity weights	Each data movement is 1 CFP

Traditional Function Points

Often focuses on business applications
Based on IFPUG approach commonly

COSMIC Function Points

Suitable for many types of software
Based on COSMIC measurement method

18. COSMIC vs LOC

LOC

Measures lines of code

Language dependent

Difficult before coding

Measures implementation size

Same function can have different LOC

COSMIC

Measures functional size

More language independent

Can be estimated earlier

Measures functionality

Functionality can be measured consistently

Example

A login function could be implemented using:

- 100 lines of Java
- 50 lines of Python
- 150 lines of C++

LOC gives different measurements.

COSMIC focuses on the **functional data movements**, not the programming language.

19. Advantages of COSMIC

1. Language Independent

It does not depend on whether software is written in:

- Java
 - Python
 - C++
 - JavaScript
 - C#
-

2. Useful for Estimation

COSMIC size can be related to:

- Effort
- Cost
- Schedule
- Productivity

For example:

If historical productivity is:

10 CFP per person-month

and the project size is:

100 CFP

Then approximate effort:

$100/10=10$ person-months

3. Useful Early in Development

You can estimate functionality before knowing the exact implementation.

4. Suitable for Different Software Types

COSMIC can be applied to many types of software, including:

- Business systems
 - Real-time systems
 - Embedded systems
 - Web applications
 - Mobile applications
-

5. Helps Compare Productivity

Suppose:

Team A:

$100 \text{ CFP} / 10 \text{ person-months} = 10 \text{ CFP/PM}$

Team B:

$100 \text{ CFP} / 20 \text{ person-months} = 5 \text{ CFP/PM}$

Team A has higher measured productivity under the same measurement basis.

20. Limitations of COSMIC

1. Requires Training

Students and new project managers need to understand the measurement rules.

2. Identification Can Be Difficult

Correctly identifying functional processes and data movements requires analysis.

3. Does Not Directly Give Cost

COSMIC gives **functional size**.

You still need productivity/cost information to calculate effort and cost.

4. Historical Data Is Useful

To convert COSMIC size into effort, organizations benefit from historical productivity data.

21. COSMIC and Software Estimation

This is especially important in **Software Project Management**.

COSMIC itself primarily gives you:

Functional Size

Then historical productivity can be used to estimate effort.

Example

Suppose:

COSMIC size = 500 CFP

Historical productivity:

25 CFP/person-month

Then:

Effort=25500 20 person-months

If average cost is:

₹1 lakh/person-month

Then:

20×₹1,00,000 ₹20 lakh

Therefore:

COSMIC Size
↓
Productivity Data
↓
Effort
↓
Labour Cost
↓
Total Project Cost

22. Important Exam Example

Question:

Calculate the COSMIC functional size for a process having:

- 2 Entries
- 1 Exit
- 2 Reads
- 1 Write

Solution:

Formula:

$CFP = E + X + R + W$

Substitute:

$CFP = 2 + 1 + 2 + 1$ CFP=6

Therefore, the functional size is **6 COSMIC Function Points**.

23. Important Terminology

Term	Meaning
COSMIC	Common Software Measurement International Consortium
CFP	COSMIC Function Point
Functional Size	Amount of functionality provided by software
Entry (E)	Data enters software from external user/system
Exit (X)	Data leaves software to external user/system
Read (R)	Software retrieves data from persistent storage
Write (W)	Software stores data in persistent storage
Functional Process	Set of activities that accomplishes a meaningful task
Software Boundary	Boundary separating measured software from external entities
